

13

D

$$pV = nRT$$

$$(2p)(2V) = nRT_2$$

$$\frac{T_2}{T} = \frac{4pV}{pV} = 4$$

Since molecular KE is proportional to temperature, the molecular KE changes 4 times too.

14

C

$$KE = \frac{1}{2}m\omega^2(x_0^2 - x^2)$$